

# Integrative analysis of bacterial diversity in patients with type 2 diabetes mellitus and periodontitis using a case-control microbiome approach

Héctor Martínez-Gregorio<sup>1,2,‡</sup>, Fernando Vallejo-Lecuona<sup>1,2,‡</sup>, Fernando Ambriz-Barrera<sup>1,2</sup>, Miguel Ruiz-De-La-Cruz<sup>1,2</sup>, Julieta Aurora Morales-Valdez<sup>1,2</sup>, Ernesto Arturo Rojas-Jiménez<sup>3</sup>, Marcela Sofía Jiménez-Martínez<sup>4</sup>, Eric Monroy-Pérez<sup>4</sup>, Gloria Luz Paniagua-Contreras<sup>4,\*</sup>, Clara Estela Díaz-Velásquez<sup>1,2,\*</sup>, Felipe Vaca-Paniagua<sup>1,2,\*</sup>

<sup>1</sup>National Health Laboratory, Molecular Diagnosis and Environmental Effects in Chronic-Degenerative Diseases, Faculty of Higher Education Iztacala, National Autonomous University of Mexico, Tlalnepantla 54090, Mexico

<sup>2</sup>Unit of Biomedicine, Faculty of Higher Education Iztacala, National Autonomous University of Mexico (UNAM), Tlalnepantla 54090, Mexico

<sup>3</sup>Department of Medical Oncology, Dana-Farber Cancer Institute, Boston, MA 02215, USA

<sup>4</sup>Faculty of Higher Education Iztacala, National Autonomous University of Mexico (UNAM), Tlalnepantla 54090, Mexico

\*Corresponding authors. Gloria Luz Paniagua-Contreras, Faculty of Higher Education Iztacala, National Autonomous University of Mexico (UNAM), Tlalnepantla 54090, Mexico. E-mail: [mya@unam.mx](mailto:mya@unam.mx); Clara Estela Díaz-Velásquez, National Health Laboratory, Molecular Diagnosis and Environmental Effects in Chronic-Degenerative Diseases, Faculty of Higher Education Iztacala, National Autonomous University of Mexico, Tlalnepantla 54090, Mexico. E-mail: [cdiazvelasquez@aol.com](mailto:cdiazvelasquez@aol.com); Felipe Vaca-Paniagua, Unit of Biomedicine, Faculty of Higher Education Iztacala, National Autonomous University of Mexico (UNAM), Tlalnepantla 54090, Mexico. E-mail: [felipe.vaca@iztacala.unam.mx](mailto:felipe.vaca@iztacala.unam.mx).

<sup>‡</sup>Contributed equally to this work

## Abstract

**Aim** Type 2 diabetes mellitus (T2DM) is a risk factor for periodontitis; however, its relationship with the oral microbiome remains poorly defined. Here, we characterize taxonomic composition and predicted microbial functions in the salivary microbiome of T2DM and non-diabetic patients with periodontitis, while evaluating the influence of disease stage, smoking, and sex.

**Methods and results** We analyzed saliva from 100 periodontitis patients (50 T2DM, 50 non-diabetic controls) using 16S rRNA V3–V4 sequencing, selecting a screened, deliberately balanced cohort to ensure robust comparison.  $\alpha$ -diversity did not differ significantly across diabetes status, disease stage, smoking, or sex. However,  $\beta$ -diversity analysis revealed a statistically significant differences in microbial community composition between T2DM and non-diabetic controls ( $R^2 = 0.016$ ,  $p = 0.029$ ). Differential abundance analysis showed enrichment of *Neisseria* ( $p = 0.02$ , Wilcoxon test with Bonferroni correction) and *Corynebacterium matruchotii* ( $p = 0.002$ ) in T2DM patients, whereas *Haemophilus sputorum* was enriched in mild periodontitis. Among smokers, *Neisseria elongata* ( $p = 0.009$ ) was more abundant than in non-smokers. Female patients exhibited higher abundances of *Haemophilus*, *Neisseria*, *Gemella*, *Aggregatibacter*, *Corynebacterium*, and *Cardiobacterium* ( $p < 0.02$ ), whereas *Megasphaera* was enriched in males ( $p = 0.04$ ). PICRUST-based predictions indicated T2DM-associated enrichment of pathway related to virulence, nutrient acquisition, and pyridoxal-5'-phosphate biosynthesis, particularly in moderate and severe periodontitis.

**Conclusions** T2DM is associated with distinct shifts in salivary microbial composition and predicted function, revealing taxa and pathways linked to periodontitis.

## Impact statement

These findings extend current evidence, supporting T2DM as an ecological modifier of the oral microbiome in periodontitis and highlighting disease-relevant microbial shifts.

**Keywords** microbiome, periodontitis, type 2 diabetes mellitus, smoking, next generation sequencing, 16S rRNA

## Introduction

Periodontal diseases, such as gingivitis and periodontitis, are chronic inflammatory conditions characterized by the progressive destruction of the periodontal complex, including the gingiva, periodontal ligaments, root cementum and alveolar bone, and affect approximately 19% of the global adult population, accounting for over 1 billion cases worldwide (Kinane et al. 2017, Łasica et al. 2024, Jain et al. 2024). Periodontal disease typically begins as gingivitis, which if left untreated, can progress to periodontitis, leading to irreversible destruction of the supporting periodontal tissues, alveolar bone resorption, and eventual tooth loss (Łasica et al. 2024, Hashim et al. 2025). Periodontitis develops through complex interactions between the oral microbiome, host immune and genetic susceptibility, behavioral factors and systemic conditions, and inadequate oral hygiene, smoking, stress, systemic disease and diabetes may contribute to periodontitis onset (Genco and Borgnakke 2013, Lertpimonchai et al. 2017, Muñoz Aguilera et al. 2020, Łasica et al. 2024, Hashim et al. 2025).

Type 2 diabetes mellitus (T2DM) is a progressive metabolic disease characterized by chronic hyperglycemia and degenerative complications, affecting more than 537 million people and has raised its prevalence by around 60% in the 2010–2021 period (Ong et al. 2023). In 2020, Mexico recorded 148 437 deaths directly attributable to diabetes, compared to an average of 101 496 deaths from 2017 to 2019, representing a 41.6% increase (Bello-Chavolla et al. 2022). T2DM has been associated with an increased risk, prevalence, and severity of periodontitis (Wu et al. 2020, Chien et al. 2023). Long-term cohort analyses indicate that adults with T2DM have nearly twice the risk of developing periodontitis compared with non-diabetic individuals (Chien et al. 2023), and poor glycemic control further exacerbates periodontal tissues destruction (Wu et al. 2020). Importantly, the coexistence of T2DM and periodontitis carries substantial clinical implications, with affected individual showing increased cerebrovascular risk and higher mortality from cardiovascular and renal disease (Saremi et al. 2005, Costa et al. 2023, Alwithanani 2023). Collectively, these complications highlight that T2DM and periodontitis are interconnected conditions with important systematic health complications.

T2DM and non-diabetic individuals with periodontitis have different bacterial compositions in their periodontal microbiomes (Shi et al. 2020a, Zhou et al. 2013, Matsha et al. 2020, Wang et al. 2023, Gu et al. 2024). In subgingival plaque, T2DM patients with periodontitis show increased abundances of *Neisseria* and *Pseudomonas*, while non-diabetic individuals exhibit reduced levels of *Corynebacterium matruchotii*, *Cardiobacterium valvarum*, and *Neisseria flavescens*, alongside higher relative abundance of *Prevotella* and *Tannerella* (Zhou et al. 2013). Consistently, plaque analysis reveal enrichment of *Fusobacteria* and *Actinobacteria* and depletion of *Proteobacteria* in T2DM subjects (Matsha et al. 2020). Other studies report higher levels of *Veillonella parvula* and *Corynebacterium matruchotii* in T2DM patients, whereas *Porphyromonas endodontalis* and *Porphyromonas gingivalis* are more abundant in non-diabetic controls (Shi et al. 2020a). Moreover, T2DM patients harbor increased proportion of *Porphyromonas*, *Fretibacterium*, *Fusobacterium*, *Rothia*, and *Filifactor* in salivary and subgingival plaque (Wang et al. 2023). Salivary microbiome

profiling further indicates stage-dependent shifts in T2DM, with predominance of *Streptococcus*, *Veillonella*, *Neisseria*, *Rothia*, and *Prevotella* across disease stages (Gu et al. 2024). Collectively, these findings support that alterations in bacterial composition in T2DM are closely linked to periodontal disease severity and progression.

Currently, the number of studies examining the oral microbiome in T2DM patients remains limited, and the taxonomic composition and functional potential of these microbial communities are still insufficiently characterized. Thus, we aimed to understand the complexity of the bacterial consortia causing periodontal disease in T2DM patients by sequencing the V3–V4 hypervariable region of the 16S rRNA gene and by analyzing the metabolic pathways implied in the ecological changes.

## Materials and methods

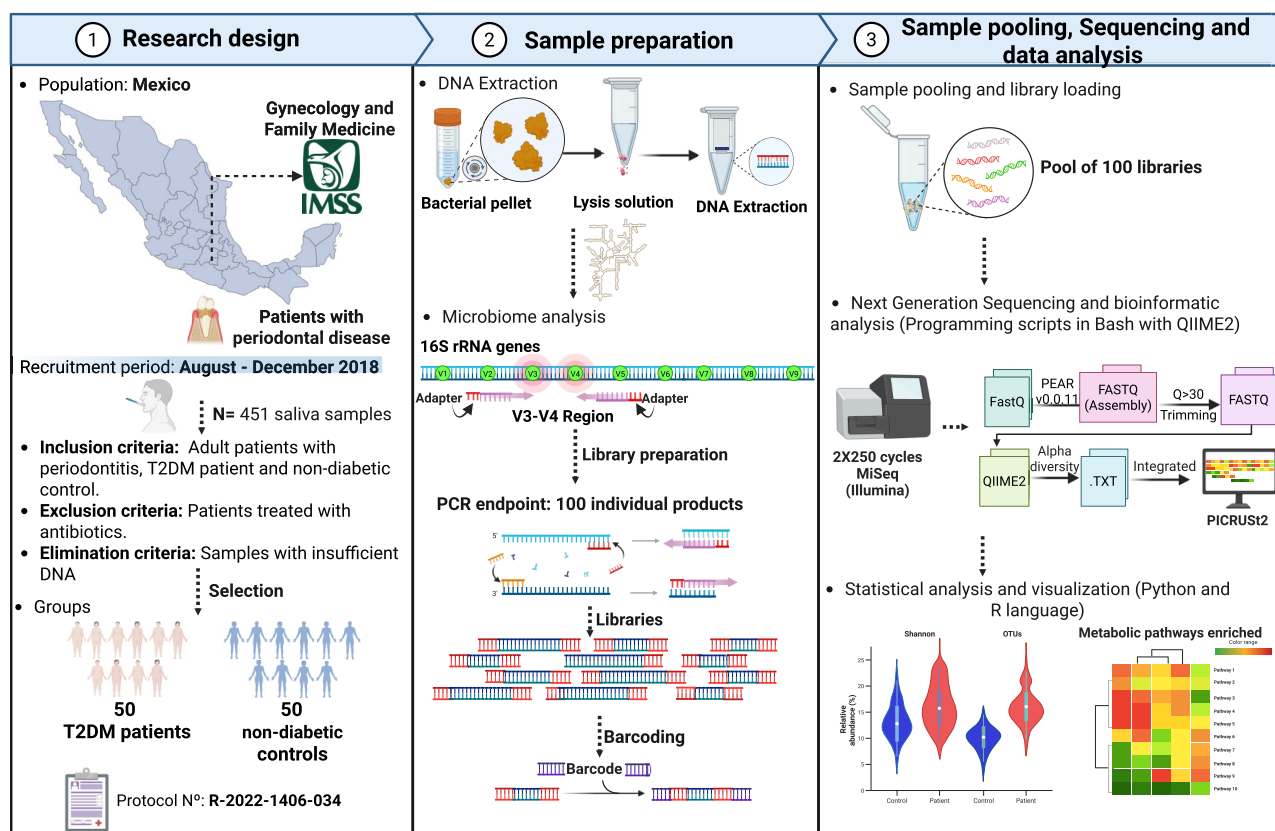
### Study population

We collected 451 saliva samples from 451 patients with periodontitis from August to December 2018 from the dental service of the Obstetrics Gynecology and Family Medicine Hospital No. 60 of the Mexican Institute of Social Security (IMSS) (Fig. 1). From this cohort, we selected 100 participants (Table S1), 50 with T2DM and 50 non-diabetic controls, who met the following criteria, a) Inclusion criteria: adult patients, IMSS beneficiaries with periodontitis over 18 years of age, with T2DM and non-diabetic controls, b) Exclusion criteria: patients treated with antibiotics in the last three months, patients with chronic-degenerative diseases, and samples with insufficient DNA for the development of the project. The final 1:1 distribution was implemented to achieve a balanced cohort and enhance analytical robustness.

The periodontal stage was determined by a certified dentist and classified according to the stage of the periodontitis as mild, moderate, and severe. Additionally, current smoking was categorized as either smokers or non-smokers. For this study, the protocol was approved by the institution's Ethics Committee (Project Identification Code: R-2022-1406-034) and conducted in accordance with the Declaration of Helsinki. All patients provided written informed consent for research purposes.

### gDNA extraction

A total of 0.5 mL of unstimulated saliva was collected from each patient in sterile Eppendorf tubes. The samples were rapidly frozen using dry ice to inhibit DNA degradation by salivary enzymes and stored at  $-80^{\circ}\text{C}$ . Then, saliva samples were processed for microbiome analysis as previously described (Janem et al. 2017). Briefly, the samples were centrifuged at  $4^{\circ}\text{C}$  for 15 minutes at  $1500 \times g$ . The pellet was resuspended in 300  $\mu\text{L}$  of lysis solution (20 mg/mL lysozyme in 20 mM Tris-HCl, pH 8.0; 2 mM EDTA; 1.2% Triton X-100) and incubated at  $37^{\circ}\text{C}$  for 1 h. gDNA extraction was performed with the DNeasy Blood & Tissue kit (Qiagen, Hilden, Germany), following the manufacturer's instructions. DNA integrity was verified on a 0.8% agarose gel and purity by spectrophotometry (Implen, Germany). Finally, DNA was quantified using Qubit (Invitrogen, Carlsbad, USA).



**Figure 1** Overview of the study workflow. (1) Experimental design and sample selection. A total of 100 patients diagnosed with periodontitis were enrolled and stratified into two groups: T2DM patients and non-diabetic controls. (2) Sample preparation. Microbial DNA was extracted, followed by amplification of the V3-V4 region of the 16S rRNA gene and subsequent barcoding. (3) Sequencing and data analysis. Next generation sequencing was performed, and data were processed using several algorithms implemented in bash, python, and R. Created with BioRender.com.

## Library preparation and massive sequencing

Amplification of the V3-V4 16S rRNA hypervariable region was carried out with forward V3 (5'TCGTCGG CAGCGTCAGATGTGTATAAGAGACAGCCTACGGGNGGCWGCAG3') and reverse V4 primers (5'GTCTCTGGGCTCGGAGATGTGTATAAG AGACAGGACTACHVGGGTATCTAATCC3') using 100 ng of gDNA. The library preparation was done following the Metagenomic Illumina 16S rRNA protocol. Paired-end sequencing was performed on an Illumina MiSeq for 2 × 250 cycles at the National Health Laboratory Molecular Diagnosis and Environmental Effect on Chronic-Degenerative Diseases, Faculty of Higher Studies Iztacala, UNAM, Mexico.

## Bioinformatics analysis

The forward and reverse FASTQ reads were assembled with PEAR v0.9.11 (Zhang et al. 2014), employing stringent quality control parameters, including a base quality cut-off of > 30 and a minimum read length of 400 bp. Microbiome analysis was done with QIIME2 v.2023.9 software incorporating the denoising function and eliminating chimeric reads (Bolyen et al. 2019). Taxonomic assignment was performed using the SILVA 132

database (Yilmaz et al. 2014). Rarefaction curves were used to verify the sequencing efficiency and the representativeness of the data. Downstream microbiome analysis was conducted in R using the microeco package v.1.15.0 (Liu et al. 2025). Operative taxonomic units (OTUs) were filtered according to the following criteria: (i) exclusion of sequences not classified as Bacteria or Archaea; (ii) removal of OTUs assigned to mitochondria or chloroplasts; (iii) exclusion of OTUs with relative abundance < 0.1%; and rarefaction of samples to 20 000 reads for data normalization.

$\alpha$ -diversity was assessed using the number of OTUs, the Shannon index, and the inverse Simpson index, while  $\beta$ -diversity was evaluated with the Bray-Curtis dissimilarity index and tested for significance using PERMANOVA. Differential abundance analysis was performed using LEfSe (Linear Discriminant Analysis Effect Size), followed by LDA, to identify the most discriminative microbial features across diabetes status, periodontitis stage, smoking, and sex. To predict functional profiles in the periodontitis stages, we used PICRUSt2 (Douglas et al. 2020) to infer the metabolic pathways enriched in the microbial communities of patients and controls. Data from the periodontitis stages were normalized using z-score transformation:  $z = (x - \mu) / \sigma$ . Visualization was performed using ggplicrust2 for pathway abundance and ComplexHeatmap for clustering and comparative analysis (Gu et al. 2016, Yang et al. 2023).

## Statistical analysis

Comparisons of categorical data between patients and controls, as well as associations between the relative abundance of bacterial taxa and clinical variables, were assessed using the chi-square test or Student's t-test for categorical variables, and the Kruskal–Wallis or Wilcoxon rank-sum test for continuous variables, followed by Yates' continuity correction or Bonferroni correction for false discovery rate (FDR). After correction, values with  $p \leq 0.05$  were considered statistically significant.

## Results

### Clinical characteristics of patients with periodontal disease

A total of 100 patients with periodontitis were selected, including 50 T2DM patients and 50 non-diabetic controls (Fig. 1). Age distribution was similar between T2DM patients and non-diabetic controls (Table 1 and Table S1). The proportion of female participants was higher in the diabetic (70%) compared to controls (52%). Smoking prevalence was low in both groups, reported in 20% of diabetic patients and 12% in controls. Severe periodontitis was the most frequently observed condition, affecting 48% of participants in each group. There was no statistical association between the epidemiological characteristics with diabetes status.

### Differences in microbiome diversity and bacterial enrichment between T2DM patients and non-diabetic controls

Quality filtering of sequencing reads was performed to remove low-quality bases, chimeric sequences, and artifacts. Overall, 30% of the reads ( $3\,829\,396 \pm 6\,264$ ) were discarded, while 70% ( $8\,946\,787 \pm 17\,267$ ) were retained for downstream. The mean reads of retained samples was 89,468X (Table S2). After filtering, one non-diabetic control sample showed insufficient sequencing depth to achieve the expected diversity plateau in the rarefaction curve and was therefore excluded (Supp. Fig. 1).

We assessed  $\alpha$ -diversity to explore variations in microbial community composition within each patient of the cohort. The  $\alpha$ -diversity (OTUs, Shannon and inverse Simpson indexes) of oral microbiota in T2DM patients was slightly higher than in non-diabetic controls; however, this difference was not statistically significant (OTUs: mean 292 vs 289,  $P = 0.99$ , Shannon index: mean 4.2 vs 4.1,  $P = 0.48$ , and Inverse Simpson Index: mean 32.5 vs 27.8,  $P = 0.47$ , respectively; Wilcoxon signed-rank test with Bonferroni correction, Fig. 2a). When clinical variables were further stratified by periodontitis stages, smoking status, and sex, no significant differences in  $\alpha$ -diversity were observed across groups ( $P > 0.05$ , Supp. Fig. 2).

The principal coordinate analysis (PCoA) based on Bray–Curtis dissimilarity was used to assess  $\beta$ -diversity and revealed significant differences in microbial community composition between T2DM patients and non-diabetic controls ( $R^2 = 0.016$ ,  $P = 0.029$ , Fig 2b). No significant differences were detected in  $\beta$ -diversity according to sex ( $R^2 = 0.015$ ,  $P = 0.056$ ), smoking status ( $R^2 = 0.014$ ,  $P = 0.07$ ) or periodontitis stage ( $R^2 = 0.025$ ,  $P = 0.07$ , Fig. Supp. 3).

LEfSe analysis followed by LDA was conducted to identify microbiome features differentially abundant across the clinical variables. Based on effect size, two genera (*Stomatobaculum* and *Eubacterium*) were enriched in the non-diabetic controls, whereas 11 taxa were enriched in T2DM patients, with *C. matruchotii* and *Porphyromonas pasteri* showing the greatest enrichment at the species level (Fig. 2c). Regarding periodontitis stages, one genus (*Candidatus\_Saccharimonas*) and two species (*Fusobacterium periodonticum* and *Haemophilus sputorum*) were enriched in the mild stage, whereas four taxa were enriched in the moderate stage, including one well-characterized *Clostridiales* species (*Clostridiales bacterium*, Fig. 2d). Among smokers, four taxa were enriched, including one species (*Lactobacillus fermentum*), whereas 18 taxa were enriched in non-smokers, five of which were identified at the species level (*Neisseria perflava*, *F. periodonticum*, *Neisseria elongata*, *Prevotella loescheii*, and *Prevotella saccharolytica*, Supp. Fig. 4a). With respect to sex, two taxa were enriched in males including one species (*Megasphaera micronuciformis*), whereas 28 taxa were enriched in females with three defined species (*C. matruchotii*, SR1 bacterium, and *Corynebacterium durum*; Supp. Fig. 4b).

### Identification of bacteria associated with T2DM patients and non-diabetic controls

Using the taxonomic information obtained from QIIME2, we identified the 10 most abundant taxa across the studied variables. At the genus level, *Streptococcus*, *Veillonella*, *Prevotella* and *Rothia* showed the highest relative abundance across T2DM patients and non-diabetic controls, as well as across periodontitis stage, smoking status, and sex. At the level species, *Streptococcus salivarius*, *Prevotella melaninogenica*, *Schaalia odontolytica* and *M. micronuciformis* represented the most abundant taxa. (Fig. 3a-b; Supp. Fig 5a-b).

Statistical analysis of microbiome composition revealed significant differences across the evaluated categorical variables at the genus and species levels. *Neisseria* ( $P = 0.02$ , Wilcoxon rank-sum test with Bonferroni correction), *Corynebacterium* ( $P = 0.01$ ), and *C. matruchotii* ( $P = 0.005$ ) showed higher relative abundance in the T2DM patients compared to non-diabetic controls (Fig. 3c), whereas *Haemophilus sputorum* ( $P = 0.002$ ) was more abundant in mild than in severe periodontitis (Fig. 3d). In females, *Haemophilus*, *Neisseria*, *Gemella*, *Aggregatibacter*, *Corynebacterium*, and *Cardiobacterium* were more abundant than in males (Wilcoxon rank-sum test,  $P < 0.02$ , Supp. Fig 5c), whereas *Megasphaera* was enriched in males ( $P = 0.04$ ). In smokers, *N. elongata* ( $P = 0.009$ ) showed higher relative abundance compared with non-smokers (Supp. Fig. 5d).

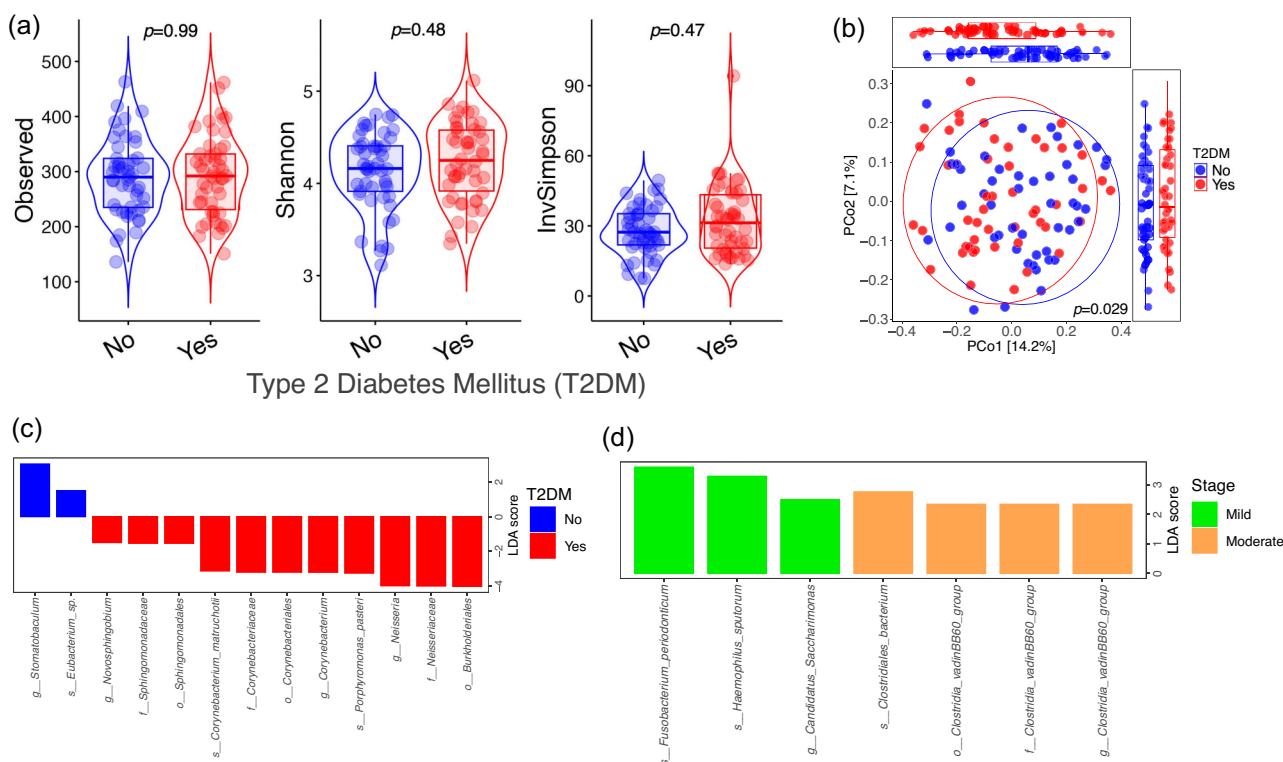
### Metabolic pathway involvement in periodontal disease

We performed a predictive analysis of the abundance of metabolic pathways and gene families associated with periodontitis stage and diabetic states using PICRUST2. Sixteen pathways were identified as statistically significant and clustered into two distinct clades: i) clade I, predominantly enriched in T2DM patients and ii) clade II, enriched in the non-diabetic controls (Fig. 4a). In mild



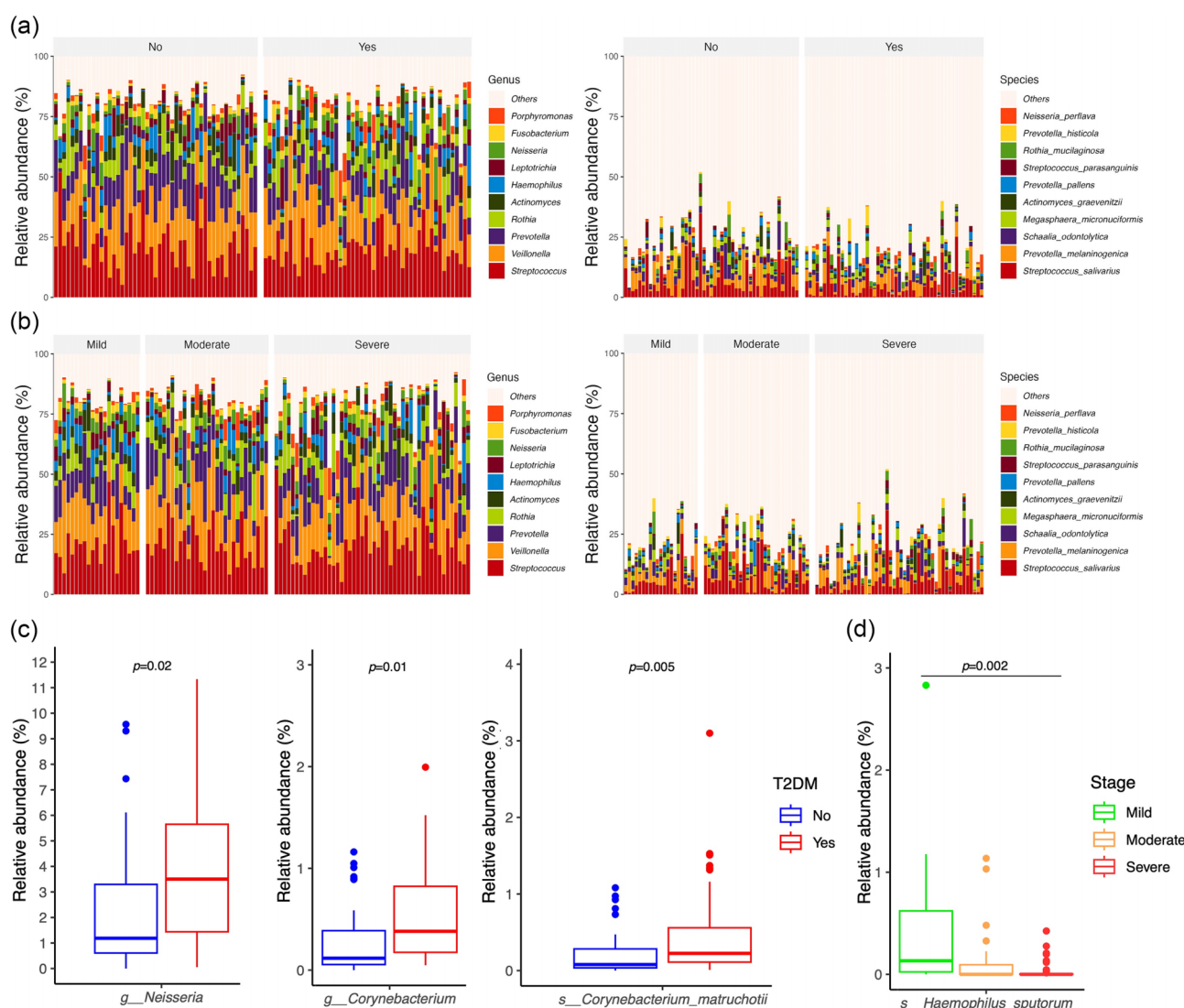
**Table 1** Epidemiological characteristics of T2DM patients and non-diabetic controls.

| Epidemiological features |          | T2DM        |           | Non-diabetic controls |           | P-value           |
|--------------------------|----------|-------------|-----------|-----------------------|-----------|-------------------|
|                          |          | N = 50<br>n | 100%<br>% | N = 50<br>n           | 100%<br>% |                   |
| Sex                      | Female   | 35          | 70        | 26                    | 52        | 0.1 <sup>a</sup>  |
|                          | Male     | 15          | 30        | 24                    | 48        |                   |
| Periodontal stage        | Mild     | 10          | 20        | 11                    | 22        | 0.96 <sup>a</sup> |
|                          | Moderate | 16          | 32        | 15                    | 30        |                   |
|                          | Severe   | 24          | 48        | 24                    | 48        |                   |
| Smokers                  | No       | 40          | 80        | 44                    | 88        | 0.4 <sup>a</sup>  |
|                          | Yes      | 10          | 20        | 6                     | 12        |                   |
|                          |          | Mean ± SD   | Range     | Mean ± SD             | Range     |                   |
| Age                      |          | 61.7 ± 10.7 | 35–86     | 59.2 ± 13.9           | 30–83     | 0.3 <sup>b</sup>  |

<sup>a</sup> Chi-square test followed by Yates' continuity correction<sup>b</sup> t-test**Figure 2** Microbiome diversity and bacterial enrichment in diabetic patients and across periodontal stages. (a) Number of OTUs per group;  $\alpha$ -diversity metrics (Shannon and inverted Simpson) in T2DM patients and non-diabetic controls. (b)  $\beta$ -diversity estimated using principal coordinate analysis (PCoA) based on Bray-Curtis dissimilarity, comparing T2DM patients versus non-diabetic controls. (c-d) LEfSe analysis showing bacterial taxa enrichment in T2DM patients versus non-diabetic controls and according to periodontitis stage, respectively.

periodontitis, pathways related to anhydromuropeptide recycling and tRNA processing were increased in relative abundance, whereas two pathways involved in pyridoxal 5'-phosphate biosynthesis and 4-aminobutanoate degradation were more prevalent in moderate and severe cases (Fig. 4b). Statistical analysis of metabolic pathways, stratified by periodontitis status, revealed significant differences across all stages of periodontitis (Fig. 4c). The superpathway of pyridoxal 5'-phosphate biosynthesis and sal-

vage showed significant associations in mild ( $P = 0.02$ , Wilcoxon rank-sum test with Bonferroni correction), moderate ( $p = 0.04$ , t-test), and severe stages ( $p = 0.003$ , Wilcoxon rank-sum test). In moderate and severe stages, significant differences were also observed in 4-aminobutanoate degradation V ( $P = 0.001$  and  $P = 1.52 \times 10^{-6}$ , t-test, respectively) and in ubiquinol biosynthesis ( $P = 0.04$  in moderate, t-test;  $P = 0.01$  in severe, Wilcoxon rank-sum test).



**Figure 3** The most abundant taxa in the microbiome according to diabetes and periodontitis status. Top 10 most abundant taxa at the genus and species levels in T2DM patients (a) and periodontitis stage (b). Taxa showing significant differences between T2DM patient and non-diabetic controls (c) and periodontitis stage (d). Statistical analysis was performed with Kruskal Wallis test or Wilcoxon ranked-sum tests with Bonferroni correction.

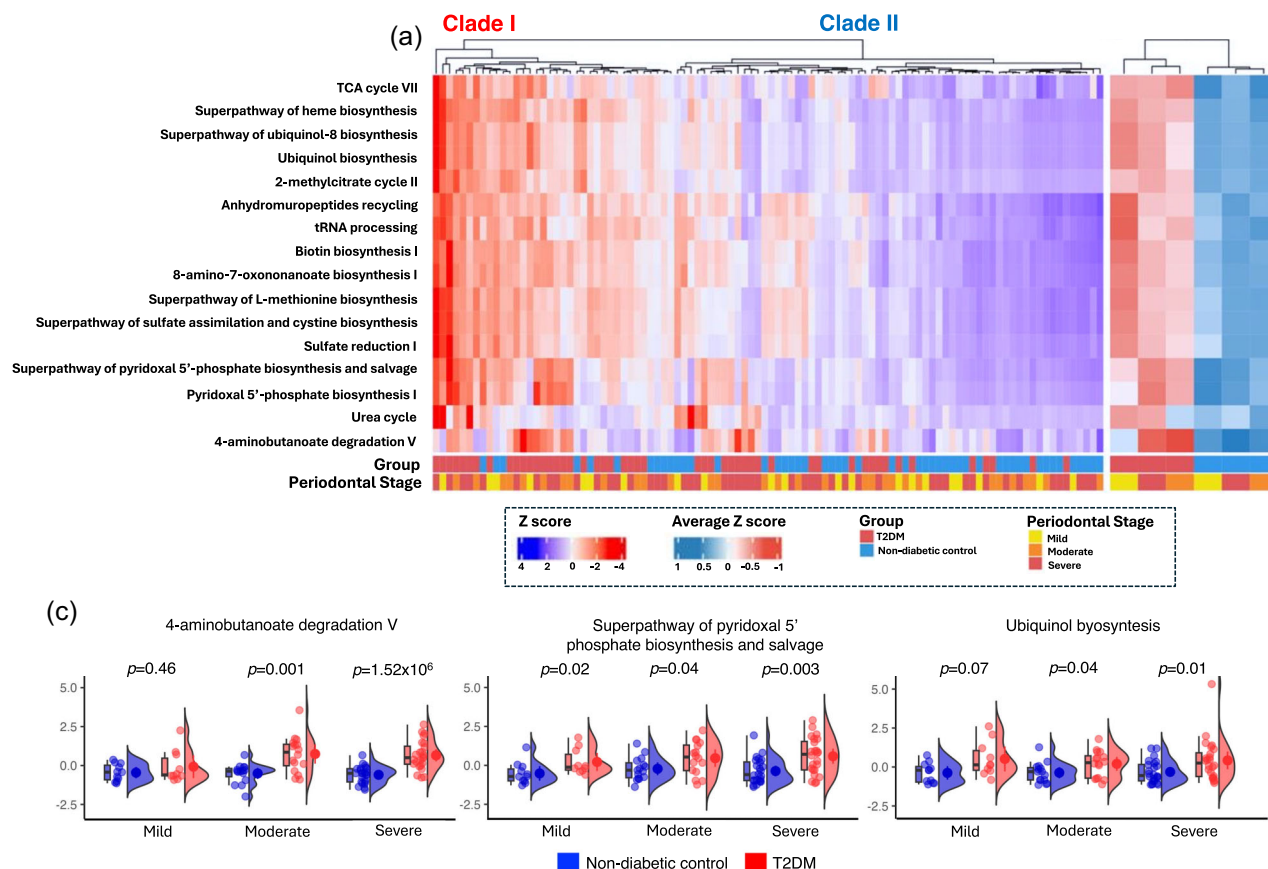
## Discussion

In this study, we characterized the oral microbiome of T2DM patients and non-diabetic controls with periodontitis using saliva-based V3-V4 16S rRNA gene sequencing as a non-invasive surrogate, examining associations with disease stage, smoking status, and sex. Understanding the ecological consequences of diabetes on the oral microbiome is critical for clarifying its contribution to periodontal disease progression and systemic health.

### Diversity patterns

We observed an increase in the  $\alpha$ -diversity in patients with T2DM compared to non-diabetic controls. However, this difference did not reach statistical significance, suggesting that diabetes may have a limited impact on within-sample microbial richness and may instead preferentially alter relative abundances. Similarly, no significant differences in  $\alpha$ -diversity were observed when stratifying by periodontitis stage, smoking status, or sex. Previous studies

have reported increased, decreased, or unchanged  $\alpha$ -diversity in diabetes, depending on the sample type, study population, and design (Ebersole et al. 2008, Zhou et al. 2013, Gu et al. 2024), and this lack of consensus is likely attributable to differences in study populations and clinical characteristics, particularly the degree of glycemic control among diabetic patients. PCoA based on Bray-Curtis dissimilarity revealed modest effect sizes but significant differences in microbial community composition between T2DM patients and non-diabetic controls ( $R^2 = 0.016$ ,  $p = 0.029$ ). Although the variance explained was small, these findings indicate consistent alterations in community structure. Notably, this pattern—characterized by stable  $\alpha$ -diversity alongside altered  $\beta$ -diversity—suggests that dysbiosis in T2DM is primarily compositional, reflecting changes in relative abundances rather than changes in overall microbial richness. No significant differences were observed by sex, smoking status, or periodontitis stage, suggesting that hormonal, immunological, or behavioral influences may contribute only modestly to salivary microbial variation in this cohort (Said et al. 2014).



**Figure 4** Metabolic pathway enrichment. (a) Comparative enrichment by diabetic status and periodontal stages. The pathways were normalized by the Z-score and arranged hierarchically. (b) Average Z-score pathway values by periodontal stage and diabetic status. Group (T2DM patients, non-diabetic controls), periodontal stage (mild, moderate, severe) are presented in the bottom rows. (c) Differential enrichment of statistically significant pathways stratified by periodontal stage in T2DM patients and non-diabetic controls.

## Microbial signatures of diabetes and periodontitis stage

LEfSe analysis revealed clear microbial signatures associated with T2DM, disease stage, smoking, and sex, underscoring their role as ecological modifiers of the oral microbiome in periodontitis. T2DM patients showed enrichment of *Corynebacterium matruchotii* and *Porphyromonas pasteri*, whereas non-diabetic controls displayed reduced abundance of *Eubacterium* and *Stomatobaculum* were observed. Increased abundance of *Corynebacterium* and *Porphyromonas* has similarly been reported in T2DM patients with periodontitis (Matsha et al. 2020). These taxa are linked to biofilm maturation and inflammation processes (Mark Welch et al. 2016). This compositional shift from commensals to bridge organisms and pro-inflammatory species may promote dysbiosis and help explain the greater severity of periodontitis in T2DM patients (Ganesan et al. 2017). Similarly, stage-specific enrichments—*H. sputorum* in mild and Clostridiales taxa in moderate periodontitis suggest dynamic microbial succession, consistent with reports that commensal diversity declines as disease progresses (Griffen et al. 2012).

## Effects of smoking and sex

Smoking and sex also imprinted distinct microbial patterns. Smokers were enriched for *L. fermentum*, whereas non-smokers harbored higher levels of *Neisseria* species. *L. fermentum* has been previously reported in smokers (Al-Marzooq et al. 2022), and increase in *Neisseria* in healthy Hispanic patients and decrease in smokers with periodontitis (Ortiz et al. 2022). These shifts likely reflect microbial adaptation to oxidative stress and align with prior evidence of tobacco-associated restructuring of *Neisseria* and *Prevotella* communities (Wu et al. 2016). Sex-based differences were very clear. In females 28 taxa, including *C. matruchotii* and *C. durum*, were enriched, while only two taxa, including *M. microneuriformis*, were enriched in males. These results suggest that taxa associated with women can confer greater microbial adaptability, while less diverse male profiles can predispose to increased pathogenicity associated with dominance (Saeb et al. 2019). These compositional differences associated with sex probably reflect the influence of sex hormones on mucosal immunity, along with behavioral and environmental factors such as diet and oral hygiene (Said et al. 2014, Liu et al. 2023). Importantly, epidemiological

studies consistently report a higher prevalence and severity of periodontitis in men compared to women (Shiau and Reynolds 2010), and our data suggest that sex-related ecological differences in the oral microbiome may contribute to this imbalance.

Statistical analyses at the genus and species levels revealed significant compositional differences in diabetic patients, with higher abundances of *Neisseria*, *Corynebacterium*, and *C. matruchotii*. These are common oral residents, not novel invaders, with *C. matruchotii* acting as a structural bridge in dental biofilms, facilitating pathogen colonization (Mark Welch et al. 2016). This suggests that diabetes may predispose the oral microbiome to dysbiosis and periodontal progression without altering overall microbial richness. *H. sputorum*, enriched in mild disease, may serve as an early-stage indicator, occupying niches later displaced in severe dysbiosis.

Sex-specific significant differences included enrichment of *Haemophilus*, *Neisseria*, *Gemella*, *Aggregatibacter*, *Corynebacterium*, and *Cardiobacterium* in women, and *Megasphaera* in men. These patterns reflect hormonal, immune, and behavioral influences (Said et al. 2014), underscoring the importance of considering sex in oral microbiome studies. Smoking also shaped specific taxa; *Lactobacillus fermentum* was enriched in smokers, whereas *Neisseria perflava*, *Neisseria elongata*, and *Fusobacterium periodonticum* predominated in non-smokers, suggesting targeted selective pressures without altering global  $\alpha$ - or  $\beta$ -diversity (Wu et al. 2016). These compositional changes may increase inflammatory potential and reduce microbial resilience, highlighting the nuanced impact of smoking on oral ecology.

## Functional alterations in diabetic microbiomes

Using PICRUST2, we identified distinct microbial functions that link the oral microbiome with periodontal disease status in T2DM patients and non-diabetic controls. Sixteen metabolic pathways were identified, each associated with various stages of periodontal disease. Notably, patients with mild periodontitis, particularly those with diabetes, exhibited an increased relative abundance of pathways involved in anhydromuropeptide recycling, tRNA processing, and biotin biosynthesis I. In contrast, diabetic patients with moderate and severe periodontitis showed elevated levels of pathways related to 4-aminobutanoate degradation V, the superpathway of pyridoxal 5'-phosphate biosynthesis and salvage, and pyridoxal 5'-phosphate biosynthesis I. These metabolic pathways linked to the oral microbiome likely promote bacterial nutrition, proliferation, and the expression of virulence factors, thereby enhancing colonization, tissue destruction, and the overall severity of periodontal disease, particularly in patients with diabetes. Additionally, other metabolic pathways associated with the subgingival microbiome in diabetic patients have been reported to favor bacterial motility, virulence factor expression, and periodontal tissue inflammation (Shi et al. 2020b). Furthermore, the relative abundance of certain metabolic pathways identified in this study, such as pyridoxal 5'-phosphate biosynthesis I and the superpathway of pyridoxal 5'-phosphate biosynthesis and salvage, have also been linked to the fecal microbiome in patients with non-alcoholic fatty liver disease (NAFLD) (Abdollahiyan et al. 2023).

## Study limitations

In this study, we analyzed the distribution and composition of the bacterial microbiome in diabetic patients using advanced sequencing technology. However, some limitations must be acknowledged. First, we analyzed saliva samples rather than subgingival periodontal plaque samples. While saliva offers a non-invasive, integrative view of the oral microbiome and there is some overlap in the detection capabilities of these two sample types, periodontal samples are generally more sensitive in detecting some bacteria (Lee et al. 2020). Periodontal bacteria primarily form biofilms, whereas saliva is enriched with planktonic bacteria, which may have influenced our results. Nevertheless, salivary microbiota can reasonably reflect subgingival plaque-derived bacteria, providing a comprehensive overview of the periodontal condition in patients with periodontitis (Kageyama et al. 2017). Additionally, changes in the microbial composition of subgingival tissues are also observed in saliva, particularly in smokers and ethanol drinkers (Park et al. 2023).

A further limitation relates to taxonomic resolution. Sequencing of the V3–V4 region of the 16S rRNA does not allow reliable species-level classification for all taxa, and in our dataset only a small proportion of features could be resolved at the species level. Consequently, our interpretations are primarily based on genus-level analyses, which nonetheless provide robust resolution for dominant and well-characterized oral bacteria and are sufficient to detect biologically meaningful differences associated with periodontitis stage, diabetes status, smoking, and sex. Other limitation is that we used PICRUST to functionally interpret the metabolic composition of the bacterial communities, which was not validated experimentally.

Another limitation of our study is the absence of an assessment of viral (such as Cytomegalovirus and Epstein-Barr virus) and protist species (*Entamoeba gingivalis* and *Trichomonas tenax*), which have been shown to influence the biology, inflammation, and progression of periodontal disease (Arduino et al. 2022, Martin-Garcia et al. 2022). In addition, our sample size, while balanced across diabetic and non-diabetic groups, is an additional limitation. Previous studies investigating the relationship between microbiota composition in periodontal disease and diabetes mellitus have a sample size ranging from 12 to 2000 (Ebersole et al. 2008, Farina et al. 2019, Sabharwal et al. 2019, Rodríguez-Hernández et al. 2019, Bolyen et al. 2019, Sun et al. 2020). Therefore, our sample size remains modest compared to large-scale epidemiological studies.

Despite these limitations, this work provides a detailed characterization of the differences in the abundance of bacterial families between diabetic and non-diabetic patients and explores their association with the stage of periodontitis, smoking, and sex. Future studies should expand the cohort size, include longitudinal sampling, incorporate virus and protist detection, and integrate host immune and metabolic parameters, as well as metagenomic analysis for a better understanding of the disease.

## Conclusions

Our study demonstrates that diabetes mellitus exerts a significant ecological impact on the salivary microbiome of patients with periodontitis, primarily by reshaping community composition at  $\beta$ -diversity level and metabolic capacities. Differential abundance



testing highlighted distinct microbial signatures, with *C. matruchoi* and *P. pasteri* enriched in diabetic patients, while controls with periodontitis showed enrichment of *Stomatobaculum* and *Eubacterium*. Microbial variation was further shaped by periodontitis stage, sex, and smoking status. Functional predictions suggested that diabetes is linked to enhanced bacterial pathways related to virulence, nutrient acquisition, and cofactor metabolism, particularly in moderate to severe periodontitis, suggesting that diabetes fosters an oral environment conducive to dysbiosis and periodontal progression. Taken together, these findings show that diabetes modulates the oral microbiome in patients with periodontitis, not by altering overall diversity, but through specific taxonomic and functional changes with potential clinical consequences.

## Author contributions

Héctor Martínez-Gregorio (Software, Visualization, Writing – original draft, Writing – review & editing), Fernando Vallejo-Lecuona (Methodology, Software, Writing – original draft), Fernando Ambriz-Barrera (Software), Miguel Ruiz-De-La-Cruz (Software, Visualization, Writing – review & editing), Julieta Aurora Morales-Valdez (Software, Writing – original draft), Ernesto Arturo Rojas-Jiménez (Software, Supervision), Marcela Sofía Jiménez-Martínez (Supervision), Eric Monroy-Pérez (Data curation, Supervision), Gloria Luz Paniagua-Contreras (Conceptualization, Data curation, Funding acquisition, Resources, Supervision), Clara Estela Díaz-Velásquez (Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision), and Felipe Vaca Paniagua (Conceptualization, Investigation, Resources, Supervision, Writing – original draft, Writing – review & editing)

## Supplementary material

Supplementary material is available at *Journal of Applied Microbiology* online.

## Conflicts of interest

None declared.

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## Data availability

The data presented in the study are deposited in the Sequence Read Archive repository, accession number PRJNA1208116.

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